

Anirudh Bocha

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WORK EXPERIENCE

Samsung Research Institute Bangalore

June 2024 - August 2024

SDE Intern

Worked on a research project focused on converting whispered audio to normal speech, with applications in voice assistants, recordings etc... Responsibilities include building and training a model by modifying a BERT-like speech model as the encoder and configuring an acoustic model from FastSpeech-TTS as the decoder and a HiFi-GAN vocoder. Utilized WTIMIT and custom datasets for model training and testing.

Timbre

January 2023 - June 2023

Software Internship

Developed an Android application for diagnosing pneumonia, tuberculosis, and COVID-19 based on user cough recordings and a machine learning model. The app collects personal details and records audio, which is sent to a FastAPI backend for analysis, returning results displayed on an output page. The frontend was implemented using Java and XML, while the backend utilized AWS DynamoDB and GraphQL.

Speech processing lab @ IIITH

November 2023 - Present

Academic project manager

Managing a team of 10 in developing various applications aligned with ongoing research in the lab, where I am also pursuing honors. Key projects included creating a multilingual data corpus collection pipeline using the MERN stack, integrating a data collection app for storing audio in Google Drive with metadata in MongoDB, and a verification app for volunteer cross-checking to ensure high-quality data. Additionally, developing a full-stack application for a stuttering diagnosis model, allowing patients to record audio, which is processed to generate a report for doctors to review and prescribe necessary actions.

PROJECTS

GredIIT - [Github](#)

Built a social media platform modeled after Reddit, incorporating key features such as user posts, comments, followers, followings, and saved posts. The platform supports an active database for real-time data management and interaction, ensuring a seamless user experience. Built using the MERN stack (MongoDB, Express.js, ReactJS, and Node.js)

YT Watch Party - [Github](#)

Developed a watch party application enabling users to create rooms for synchronized viewing of YouTube videos or playlists. Integrated a chat interface for user interaction, with admin privileges allowing control over room management and admin delegation. Implemented video synchronization and chat functionality using sockets. The entire application was built with ReactJS and NodeJS.

Clan databases - [Github](#)

This is a database for Clash of Clans players to assist clan leaders in recruiting members and tracking leaderboard positions. The project was implemented using object-oriented programming (OOP) concepts in Python, with the database built in SQL. Included unit tests to ensure functionality.

Indoor air pollution dashboard - [Github](#)

Created as a part of project in a coursework, this is a real-time dashboard for an environment sensor module that monitors pollution index, temperature, humidity, and CO2 content. The data upload code, written in MATLAB and embedded in the sensor module, sends data to ThingSpeak. The dashboard retrieves data via the ThingSpeak API and displays real-time updates and graphical visualizations, built with ReactJS and NodeJS.

EDUCATION

Bachelor of Computer Science in International Institute of Information and Technology , Hyderabad.

Currently Pursuing Honors in Speech Processing - LTRC IIITH

2021 - 2025